

Akash Mahajan

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EXPERIENCE

Member of Technical Staff / Tech Lead Manager | [Contextual AI](#)

2024-2025

- System development (0→1)
 - Designed core [multimodal document understanding](#) (parsing + ETL index) used by retrieval agents for all customers in production
 - Ingesting O(billion) tokens of multimodal documents with complex layouts, tables, graphics, metadata
 - Shipped as a distributed async service with multiple stages (CPU, GPU and VLM API). I owned system quality and core modules, and co-led architecture with a platform lead
- Applied research: harness and eval design, token-efficient retrieval
 - Built core harness combining parsing tools, custom segmentation models and VLM/LLM modules for auditable OCR with bbox provenance and minimal hallucination
 - Developed composable ingest-time ETL workflows enabling efficient retrieval beyond point queries (à la [llm-wiki](#)) [[Blog: Token-efficient GraphRAG](#)] [[Demo: 250 page PDF chat in Cursor](#)]
- Tech Lead Manager
 - PoC with Product/GTM/Marketing across company; Interviewed candidates, led team of 3
 - Critical in landing company's first multi-million \$ enterprise [contract with Qualcomm](#)

Senior Applied Scientist | [Microsoft Azure Speech](#)

2018-2023

- Model development: state-of-art transcription designed for scale [O(10 million) hrs/mo]
 - Shipped both batch and streaming models to [Azure Batch](#), [Microsoft Word](#), and [Microsoft Teams](#)
 - In particular: Conformer batch model ([Whisper](#)-comparable) optimized to run at 50x realtime
- Applied research: diarized multi-speaker multi-mic transcription
 - Shipped diarized in-conference room transcription device covered by [The Verge](#)
 - Lead contributor: ASR training recipes, evaluation metrics, cross-system error analysis
- Research engineering: data pipelines, distributed training, inference
 - Sped up O(1e20) FLOP training on low-cost V100 GPUs to run in <1 week
 - Fixed inference bottlenecks leveraging NVIDIA/ONNX profiling tools, saving \$ millions

PROJECTS

Self-study: efficient audio generation, multi-stream dialog models | [\[github\]](#) (ongoing)

2026

- Speculative decoding with relaxed acceptance on single-codebook TTS model (Chatterbox-TTS)
- Meanflow distillation of FSQ token flow decoder for 2x faster local inference (Voxtral-TTS)
- Hybrid cascaded/duplex voice agent architectures (e.g. MoshiRAG)

tinydiarize: Lightweight extension of Whisper for diarization | [\[github\]](#)

2023

- Built extension of OpenAI's Whisper ASR model for speaker diarization with special tokens
- Released with [integration in whisper.cpp](#) (50k+ stars) — runnable on MacBooks/iPhones

EDUCATION

Stanford University, M.S. Management Science & Engineering

2016-2018

- Course Assistant for Machine Learning (CS229) & Deep Learning (CS230)
- Built low-cost conv-seq2seq TTS trainable with \$75 of compute; awarded [best poster](#) in [CS224n](#)

Indian Institute of Technology (IIT), Madras, B.Tech.

2011-2015

Chemical Engineering, minor: Control Systems (signal processing, stats, linear algebra)

Patents

- [US11044287B1](#): Network resilient real-time voice communication leveraging on-device speech models (granted during pandemic) Microsoft, 2021
- [WO2018020475A1](#): Electric vehicle drivetrain predictive health monitoring Ather Energy, 2018

Tech Stacks

- Contextual AI: Python/pydantic, HF Transformers, Gradio, vLLM, asyncio/Temporal
- Microsoft: Pytorch distributed, Azure ML/blob pipelines, ONNX/C++

Other

- Peer reviewer for ICASSP conference, TMLR journal, and NeurIPS, CVPR workshops 2025/26
 - Interviewed in [Mongabay News](#) for [OrcaHello](#): AI-assisted 24x7 hydrophone monitoring 2020/26
 - Organized [Chennai's largest EDM gig](#) (5k+ attendees) at IIT Madras 2014
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